

DEBI PRASAD PRADHAN

STAFF SOFTWARE ENGINEER | AI PLATFORM ENGINEER | DISTRIBUTED SYSTEMS

dpp2017@gmail.com +91-7337491586 Bangalore, India www.linkedin.com/in/debiprasadpradhan

Staff Software Engineer with 11+ years of experience building enterprise AI platforms, distributed systems, and cloud-native applications. Experienced in developing GenAI-powered products using OpenAI, LangGraph, MCP, and knowledge graphs, while designing scalable platforms on AWS. Proven track record of leading technical execution, building intelligent developer platforms, and delivering enterprise-grade AI solutions that improve engineering productivity and business outcomes.

Executive Highlights

- Designed and delivered enterprise AI platforms leveraging OpenAI, Claude, Gemini, LangGraph, LangChain, MCP, Langfuse, RAG, and Knowledge Graphs to accelerate engineering productivity and intelligent workflow automation.
- Architected AI-powered Specification-Driven Development (Workflow Forge), reducing SDLC effort by 67% and accelerating engineering delivery by 3×.
- Led technical execution across 3 engineering teams (20+ engineers), partnering with Principal Architects to deliver scalable enterprise AI and cloud-native platforms from concept to production.
- Designed AWS-based distributed systems supporting enterprise-scale scientific platforms, high-volume data processing, and large-scale file management with a focus on scalability, reliability, and cost optimization.
- Built enterprise AI, cloud-native, and platform engineering solutions emphasizing performance, resilience, security, and operational excellence.
- Established engineering standards, reusable AI platform components, architecture governance, and best practices across multiple engineering teams.
- Led modernization initiatives by reducing technical debt and evolving enterprise platforms toward cloud-native and AI-native engineering practices.
- Delivered enterprise products end-to-end, consistently translating business requirements into customer-focused, production-ready solutions.

Architecture, Leadership & Engineering Expertise

- AI Platform Engineering
 - AI Platform Development | AI Workflow Orchestration
 - Specification-Driven Development | Enterprise AI Applications | Intelligent Automation
- Distributed Systems & Platform Engineering
 - Distributed Systems | Platform Engineering | Cloud-Native Architecture
 - Microservices | API & Integration Architecture | Event-Driven Architecture
 - High Availability | Scalability | Observability | Performance Engineering
- Cloud Engineering
 - AWS | Kubernetes | Cloud Cost Optimisation (FinOps) | Enterprise Platform Architecture
- Leadership
 - Technical Leadership | Architecture Governance | Engineering Standards
 - Technical Roadmaps | Product Strategy | Customer Success | Technical Mentoring

PROFESSIONAL EXPERIENCE

Staff Software Engineer | Technical Architect

THERMO FISHER SCIENTIFIC | Bangalore, India

April 2022 - Present

Partner with the Principal Architect to define enterprise architecture, drive technical strategy, and lead technical execution across three engineering teams delivering enterprise platforms, cloud-native systems, GenAI solutions, and scientific data platforms. Responsible for architecture, engineering governance, cloud modernization, performance optimization, AWS solution design, engineering leadership, and successful delivery of enterprise-scale products.

AI Platform Engineering

- Designed and delivered enterprise AI platforms using OpenAI, Claude, Gemini, LangGraph, MCP, RAG, and Knowledge Graphs to automate engineering workflows and improve developer productivity.
- Architected Specification-Driven Development (Workflow Forge), enabling AI-assisted software engineering, reusable implementation specifications, and reducing SDLC effort by 67%.
- Developed PyWizard, an enterprise AI assistant integrating multiple LLM providers, agentic workflows, and intelligent development capabilities.
- Designed enterprise Knowledge Graph and semantic search solutions using GraphDB, SBERT, vector search, and RAG for intelligent product discovery and engineering insights.
- Built AI-powered workflow automation and enterprise knowledge systems supporting scalable engineering platforms.

Distributed Systems & Cloud Engineering

- Architected cloud-native distributed systems emphasizing scalability, resilience, observability, and performance.
- Designed AWS-native platforms using Kubernetes, Aurora PostgreSQL, RabbitMQ, GraphDB and event-driven architectures.
- Built enterprise file management platform supporting secure upload and lifecycle management of files up to 1 TB.
- Improved platform performance through distributed system design, asynchronous processing, caching, and scalable data access.
- Led cloud modernization and FinOps initiatives improving scalability, operational efficiency, and cloud cost optimization.

Technical Leadership & Engineering Excellence

- Led technical execution across three engineering teams (20+ engineers) while partnering with Principal Architects.
- Established engineering standards, reusable frameworks, architecture governance, and implementation best practices.
- Collaborated with Product Managers, UX teams, Architects, and business stakeholders to define technical strategy and platform roadmaps.
- Mentored engineers through architecture reviews, technical design sessions, and engineering best practices.
- Delivered enterprise products from concept to production while maintaining engineering quality and customer commitments.

EDUCATION

- Master of Computer Applications (MCA) — Utkal University
- Bachelor of Computer Applications (BCA) — Utkal University

SELECTED ENTERPRISE INITIATIVES

Workflow Forge - AI-Powered Specification-Driven Development Orchestrator Platform

- Architected an AI-powered Specification-Driven Development platform transforming business specifications into implementation, testing, validation, documentation, and deployment workflows.
- Designed distributed workflow orchestration using Temporal integrated with Jira and GitHub.
- Reduced SDLC effort by 67% while accelerating engineering delivery by 3x through AI-assisted software engineering.

Analytical Data Service - Enterprise Scientific Knowledge Graph Platform

- Architected an enterprise Knowledge Graph platform using GraphDB, Apache NiFi, RabbitMQ, and Python ETL for scientific data management.
- Implemented RAG-enabled semantic search using SBERT embeddings, vector search, and OpenAI to improve scientific discovery.

PyWizard - Enterprise GenAI Scientific Intelligence Platform

- Architected a multi-agent enterprise GenAI platform using LangGraph, OpenAI, Claude, Gemini, MCP, and Langfuse.
- Implemented provenance-aware AI validation, human-in-the-loop review, and intelligent workflow automation.
- Reduced scientific model preparation from weeks to hours through AI-assisted document understanding.

AI & ENGINEERING EXPERTISE

AI & Intelligent Systems - Generative AI | Agentic AI | AI Agents | LLM Integration | OpenAI | Claude | Gemini | LangGraph | MCP | RAG | Knowledge Graphs | Semantic Search | Vector Search | Embeddings

AI Platform Engineering - AI Platform Development | AI Workflow Orchestration | Specification-Driven Development | Enterprise AI Applications | Intelligent Automation

Distributed Systems & Platform Engineering - Distributed Systems | Platform Engineering | Cloud-Native Architecture | Microservices | Event-Driven Architecture | API & Integration Architecture | High Availability | Scalability | Observability | Performance Engineering

Cloud Engineering - AWS | Kubernetes | Docker | Temporal | Cloud Cost Optimization (FinOps)

Data & Knowledge Platforms - GraphDB | Neo4j | PostgreSQL | Aurora PostgreSQL | Apache NiFi | RabbitMQ | Kafka | MongoDB

Programming & Frameworks - Java | Python | SQL | Spring Boot | Spring Cloud | FastAPI

PREVIOUS EXPERIENCE

Oracle | Applications Engineer | 2019–2022

- Modernized Oracle Fusion Procurement applications by migrating Oracle ADF/JSF to Oracle JET.
- Designed REST APIs supporting Oracle Fusion ERP workflows.

ComakeIT | Software Engineer | 2017–2019

- Designed RESTful microservices using Java and Spring Boot.
- Built automated testing frameworks using Selenium, Cucumber, and JUnit.

HCL Technologies | Software Engineer | 2015–2017

- Developed enterprise applications using Java, Spring, REST, SOAP, JPA, and JDBC.
- Improved data processing performance through multithreaded programming.